

The CNN Challenge: Report

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Problem Statement: Task provided is to train and optimize a CNN to have high accuracy on a dataset provided of approx 2400 samples spread across 16 classes.

Repo link: <https://github.com/osavlani/ITCS6169-A1>

Starting with the starter notebook the CNN had an accuracy of **54.32%** while employing Grayscale, normalization, Flatten, Adam optimizer, batch size: 64 with epochs 20. Compared to traditionally deployed models we can see there's a lack of layers, augmentation, regularization, etc so less information is being processed/being learnt per epoch. For starters we employ ResNet18 as our architecture which has 11M parameters. There were many other traditional options such as AlexNet 60M, VGG16 138M, ResNet34 21M, ResNet50 25M, DenseNet121 8M, EfficientNet-B0 5M, MobileNetV3 5M, etc but ResNet18 was chosen as it has high validation accuracy on transfer learning and is lightweight to run.

Comparison of starting CNN to ResNet18

	Starter Notebook	Base ResNet18
Convolution Layers	1	18 with residual connections
Input	64x64	224x224 rgb
Augmentation	Grayscale (loses color information)	Better options can be handpicked based on dataset
Optimizer	Adam with fixed Lr	AdamW or SGD momentum with differential Lr
Regularization	None	Weight decay, dropout, etc
Skip connections	None	Every block

The basic comparison shows lack of depth in starter CNN (T-net) and input having low resolution while lacking color information. Learning rate is fixed, which can hinder accuracy as the same rate is applied on all layers, while also lacking regularization. ResNet18 has more layers which leads to higher accuracy, it takes higher resolution rgb images which contain more information to learn so that the model can estimate better on how to identify a sample with possible respective classes.

To employ ResNet18, we use Transfer Learning techniques and use pretrained weights of ResNet18 on imagenet_1K with the architecture to train on our dataset. With the model having its weights fine tuned to data pertaining to our task, the model is then validated to see its

performance. Doing so we got the validation accuracy of **94.90%**, with RGB 224px input, ImageNet normalization, augmentation such as crop/flip/jitter, 2-phase training.

Now for further optimization, we do some changes in training the ResNet18 weights. The trained resnet18 weights are treated as a checkpoint on which the model further implements training. We use cutmix/mixup augmentation which are modern augmentations proven to help gain better training metrics over existing augmentations. We also use label smoothing which lets the model store its weights as soft targets (0.1,0.3,0.4,0.2) instead of hard targets (0,0,1,0) and also uses early stopping (with patience) so that the model isn't overtrained. We also use a different optimizer which is SGD + momentum + cosine schedule over AdamW. This is done because adam optimizers help weights converge aggressively but since we are using pre-trained weights and have a small dataset SGD optimizer is better as it can let weights generalize. If we were training a new model on good amounts of data Adam optimizer could be a better choice. Also, existing literature has proven that SGD with momentum (speeds convergence with updating gradient averages) and cosine schedule (decreasing learning rate so that features are learnt early on and fine-tuned in the end) is the best deployment of SGD. Doing so we got a validation accuracy of **94.90%**.

For our final experiment we do hyper parameter tuning. As our deployed architecture shows promise but the previous optimizations underperformed compared to the expected increase in accuracy, we can tune the hyper parameters to try and get more accuracy out of the current deployment. Parameters targeted are learning rate, epochs, mixing strength(cutmix) and dropout regularization is added. We do probe runs of 5 epochs each for different parameter values to find out that learning rate is best at 0.01, mixing strength is best at 1.0, and dropout performs well at 0.3. With these parameters we do a final run of 70 epochs with early stopping put as 10. Doing so we get an accuracy of **95.79%**. 70 epochs was chosen at random, a higher number than the original 30 to check if the model learns more features if given more runs, and based on the run as epochs stopped at 43 due to early stopping we didn't need to go higher than 70.

In conclusion, having an Initialized ImageNet helped, the model was overfitting to the data, one clear indication to that was early stopping at only 30 epochs. Having an optimal optimizer helped push the accuracy by 2%. Heavier augmentation provided better accuracy in case of cut-mix, and with testing more variables such as model capacity, weight decay, input image resolution we can find further ways to optimize and get accuracy to be even higher. AI being my assistant helped me to choose from existing architectures based on estimated validation accuracy and total parameters in a model (light model preferred as it was being run locally). AI also helped me choose between possible options of optimizers that I could've employed.

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